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Integrated antenna package structure and manufacturing method thereof

Abstract

Provided is an integrated antenna package structure including a chip, a circuit structure, a shielding body, an encapsulant, a first antenna layer, a dielectric body, and a second antenna layer. The circuit structure is electrically connected to the chip. The shielding body is disposed on the circuit structure and has an accommodating space. The chip is disposed in the accommodating space of the shielding body. The encapsulant is disposed on the circuit structure and covers the chip. The first antenna layer is disposed on the circuit structure and is electrically connected to the circuit structure. The dielectric body is disposed on the encapsulant. The second antenna layer is disposed on the dielectric body. A manufacturing method of the integrated antenna package structure is also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application no. 110147083, filed on Dec. 16, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a package structure and a manufacturing method thereof, particularly to an integrated antenna package structure and a manufacturing method thereof.

Description of Related Art
(3) With the advancement of technology, electronic products are packed with more functionalities than ever. For example, a mobile communication device nowadays is provided with various electronic components with different functions that are each small in size to trend toward lightness and thinness.
(4) In the existing electronic assembly, the antenna is separated from the chip package structure, and the antenna needs to be electrically connected to the chip in the package structure through the circuit on the circuit board, which makes it difficult to reduce the overall volume of the electronic assembly. In this light, it is an urgent task to solve this problem.
SUMMARY
(5) The disclosure is directed to an integrated antenna package structure with small volume and/or good quality and a manufacturing method thereof.
(6) According to an embodiment of the disclosure, an integrated antenna package structure includes a chip, a circuit structure, a shielding body, an encapsulant, a first antenna layer, a dielectric body, and a second antenna layer. The circuit structure is electrically connected to the chip. The shielding body is disposed on the circuit structure and has an accommodating space. The chip is disposed in the accommodating space of the shielding body. The encapsulant is disposed on the circuit structure and covers the chip. The first antenna layer is disposed on the circuit structure and is electrically connected to the circuit structure. The dielectric body is disposed on the encapsulant. The second antenna layer is on the dielectric body.
(7) According to an embodiment of the disclosure, a manufacturing method of an integrated antenna package structure includes the following processes. A circuit structure is formed. A chip is disposed on the circuit structure to be electrically connected with the circuit structure. A shielding body is formed on the circuit structure, in which the shielding body has an accommodating space, and the chip is disposed in the accommodating space of the shielding body. An encapsulant is formed on the circuit structure to cover the chip. A first antenna layer is formed on the circuit structure to be electrically connected with the circuit structure. A dielectric body is formed on the encapsulant. And a second antenna layer is formed on the dielectric body.
(8) Based on the above, the integrated antenna package structure and the manufacturing method thereof of the disclosure have a smaller volume and/or better quality.

Description

BRIEF DESCRIPTION OF THE DRAWING
(1) FIG. **1**A to FIG. **1**F are partial cross-sectional schematic diagrams of part of a manufacturing method of an integrated antenna package structure according to a first embodiment of the disclosure.
(2) FIG. **1**G is a schematic partial top view of the integrated antenna package structure according to the first embodiment of the disclosure.
(3) FIG. **2** is a partial cross-sectional schematic diagram of part of a manufacturing method of an integrated antenna package structure according to a second embodiment of the disclosure.
(4) FIG. **3** is a partial cross-sectional schematic diagram of part of a manufacturing method of an integrated antenna package structure according to a third embodiment of the disclosure.
DESCRIPTION OF THE EMBODIMENTS
(5) Reference is made hereinafter in detail to the embodiments of the disclosure, examples of which are illustrated in the drawings. Wherever possible, the same reference numbers in the drawings and description refer to the same or similar parts.
(6) Unless expressly stated otherwise, directional terms (e.g., top, down, left, right, front, back, top, and bottom) herein are for reference only for the drawings and are not intended to imply absolute

orientation.
(7) Unless explicitly stated otherwise, any process described herein shall not to be construed as to be performed in a particular order.
(8) Despite that the disclosure is more fully explained with reference to the drawings of the embodiments, the disclosure can be embodied in various forms and should not be limited to the embodiments described herein. The thicknesses, sizes, or dimensions of layers or areas in the drawings may be exaggerated for clarity.
(9) FIG. **1**A to FIG. **1**F are partial cross-sectional schematic views of part of a manufacturing method of a chip package structure according to a first embodiment of the disclosure.
(10) As shown in FIG. **1**A, a circuit structure **120** is formed on a carrier **91**. There is no particular limitation on the carrier **91** in the disclosure, as long as the carrier **91** is suitable for carrying the film layer formed thereon or the components disposed thereon.
(11) In one embodiment, the carrier **91** has a release layer **92**, but the disclosure is not limited thereto. The release layer **92** includes a light-to-heat conversion (LTHC) adhesive layer or other film layers suitable.
(12) In this embodiment, the circuit structure **120** includes an insulating layer **122** and a conductive layer **121**. The conductive layer **121** constitutes corresponding circuits. The layout design of the circuit may be adjusted according to requirements, which is not limited in the disclosure. For example, in the circuits of the circuit structure **120**, the parts that are not connected in the drawings are electrically connected by other elements not shown and/or other conductive components.
(13) In this embodiment, as the circuit structure **120** is formed by common semiconductor processes (e.g., deposition process, photolithography process, and/or etching process), it is not repeated herein.
(14) In one embodiment, the circuit structure **120** may be referred to as a redistribution circuit structure, but the disclosure is not limited thereto.
(15) In one embodiment, the topmost conductive layer **123** (i.e., the layer of the conductive layer **121** farthest from the carrier **91**) is disposed on the topmost insulating layer **122**, and part of the topmost conductive layer **123** is embedded in the topmost insulating layer **122**.
(16) In an embodiment not shown, the topmost insulating layer **122** has a plurality of openings, and the openings expose the conductive layer **121** farthest from the carrier **91**.
(17) As shown in FIG. **1**A, the conductive member **140** is formed on the circuit structure **120**. At least part of the conductive member **140** is electrically connected to a corresponding part of the topmost conductive layer **123**. The conductive member **140** includes a pillar-shaped conductive member **142** and a frame-shaped conductive member **134**. In one embodiment, the frame-shaped conductive member **134** includes a frame-shaped area and other areas extending outward from the frame-shaped area. In other words, as long as the frame-shaped conductive member **134** has an area for an object to be placed therein (such as the chip **110** described in the subsequent FIG. **1**B, but it is not limited thereto), and the disclosure does not limit the appearance of the frame-shaped conductive member **134**.
(18) In one embodiment, the conductive member **140** includes a pre-formed conductive member. For example, the conductive member **140** includes a pre-formed conductive pillar or a pre-formed conductive frame, but the disclosure is not limited thereto.
(19) In one embodiment, the conductive member **140** is formed by common semiconductor processes (e.g., a photolithography process, a sputtering process, an electroplating process, and/or an etching process), but the disclosure is not limited thereto. In one embodiment, the conductive member **140** includes a plating core layer and a seed layer, but the disclosure is not limited thereto. The seed layer is disposed between the topmost conductive layer **123** and the plating core layer. In one embodiment, the seed layer surrounds the plating core layer, but the disclosure is not limited thereto.
(20) As shown in FIG. **1**B, the chip **110** is disposed on the circuit structure **120**. There is a

component area (not shown) on one side of the chip **110**, and the surface on which the component area is disposed may be referred to as an active surface **110***a*, and the other surface opposite to the active surface **110***a* may be referred to as a backside **110***b*. The chip **110** is disposed on the circuit structure **120** with its active surface **110***a* facing the circuit structure **120**. And the components in the chip **110** are electrically connected with the corresponding circuits in the circuit structure **120**.
(21) In this embodiment, the conductive member **140** is formed first, as shown in FIG. **1**A, before the chip **110** is disposed, as shown in FIG. **1**B.
(22) In an embodiment not shown, the chip **110** is configured first, before the conductive member **140** is formed.
(23) Notably, the backside **110***b* of the chip **110** may be considered part of the chip **110** if there are suitable films, layers, and/or components on the backside **110***b* of the chip **110** prior to disposing the chip **110** on the circuit structure **120**.
(24) As shown in FIG. **1**C, after the conductive member **140** is formed and the chip **110** is disposed, an encapsulant **174** is formed on the carrier **91**. The encapsulant **174** directly covers at least the backside **110***b* of the chip **110**, and the encapsulant **174** exposes the conductive member **140**.
(25) The encapsulant **174** has a first sealing surface **174***a* and a second sealing surface **174***b*. The second sealing surface **174***b* is opposite to the first sealing surface **174***a*. The first sealing surface **174***a* of the encapsulant **174** faces the circuit structure **120**. The encapsulant **174** may completely cover the entire backside **110***b* and the entire side surface **110***c* of the chip **110**, but the disclosure is not limited thereto.
(26) For example, a molding material is formed on the carrier **91**. After the molding material is cured, a planarization process may be performed. And after the planarization process is performed, the conductive member **140** may be exposed from the encapsulant **174**. In other words, the second sealing surface **174***b* of the encapsulant **174** may be coplanar with the upper surface **140***b* of the conductive member **140** (i.e., the surface of the conductive member **140** farthest from the circuit structure **120**). During the planarization process, a portion of the cured molding material and/or a portion of the conductive members **140** may be slightly removed.
(27) In this embodiment, part of the encapsulant **174** is disposed between the chip **110** and the circuit structure **120**. For example, the chip **110** is electrically connected to the corresponding circuit in the circuit structure **120** through corresponding conductive bumps **112**, and part of the encapsulant **174** disposed between the chip **110** and the circuit structure **120** may cover the conductive bumps **112**. That is, it is possible that the first sealing surface **174***a* of the encapsulant **174** is not coplanar with the active surface **110***a* of the chip **110**, the contact pad **111** (e.g., die pad) on the chip **110**, and/or the passivation layer **113** on the chip **110**, but the disclosure is not limited thereto.
(28) As shown in FIG. **1**C, in this embodiment, after the encapsulant **174** is formed, the conductive layer **150** is formed on the second sealing surface **174***b* of the encapsulant **174**. The conductive layer **150** includes a first antenna layer **156** and a conductive portion **135** different from the first antenna layer **156** aforementioned. The conductive layer **150** is formed by common semiconductor processes (e.g., deposition process, photolithography process and/or etching process; or screen printing), which are not limited in the disclosure.
(29) The antenna pattern of the first antenna layer **156** may be adjusted according to design requirements, which is not limited in the disclosure.
(30) The conductive portion **135** (which is also known as the second conductive portion **135**) and the frame-shaped conductive member **134** (which is also known as the first conductive portion **134**) are electrically connected (or even in direct contact), so as to constitute the shielding body **130**.
(31) In this embodiment, the bottom surface of the conductive layer **150**, the second sealing surface **174***b* of the encapsulant **174**, and the upper surface **140***b* of the conductive member **140** may be coplanar. In other words, the conductive layer **150** directly contacts a corresponding part of the encapsulant **174** and/or the corresponding conductive member **140**. That is, in a direction

perpendicular to the first sealing surface **174***a* or the second sealing surface **174***b*, there is basically no other substance between the conductive layer **150** and the encapsulant **174**, and/or there is basically no other substance between the conductive layer **150** and the conductive member **140**. As such, the process is relatively simple accordingly.
(32) In this embodiment, in a direction perpendicular to the first sealing surface **174***a* or the second sealing surface **174***b*, there is only the encapsulant **174** between the entire backside **110***b* of the chip **110** and the second conductive portion **135**. With the above process, the part where the encapsulant **174** is between the chip **110** and the second conductive portion **135** is able to provide sufficient protection and/or separation, which allows other substances or film layers (e.g., Die Attach Film (DAF)) between the chip **110** and the second conductive portion **135** to be omitted, and the thickness of the structure may be reduced accordingly; and/or, the process may be relatively simple.
(33) As shown in FIG. **1**D, after the encapsulant **174** and the conductive layer **150** are formed, a dielectric body **177** is formed on the encapsulant **174**.
(34) In this embodiment, a semi-cured dielectric material is coated on the encapsulant **174**. Also, after the dielectric material is cured, a corresponding dielectric body **177** is formed. The dielectric material includes, but is not limited to, a polymer material (e.g., polyimide (PI)).
(35) In an embodiment not shown, a cured or solid dielectric material (including, but not limited to, glass, polymer plate, or polymer film) is disposed on the encapsulant **174**. The cured or solid dielectric material and the encapsulant **174** may be connected to each other by an insulating adhesive material to form the dielectric body **177** correspondingly.
(36) Furthermore, in FIG. **1**D, after the dielectric body **177** is formed, a conductive layer **160** is formed on the dielectric body **177**. The conductive layer **160** at least includes the second antenna layer **166**. The conductive layer **160** is formed by common semiconductor processes (such as deposition process, photolithography process, and/or etching process; or screen printing), which is not limited in the disclosure.
(37) In this embodiment, the conductive layer **160** includes a second antenna layer **166** and a shielding layer **165**.
(38) As shown in FIG. **1**E, after the conductive layer **160** is formed, the carrier **91** may be removed.
(39) Please refer to FIG. **1**E to FIG. **1**F. A singulation process/dicing process or other suitable process may be performed on the structure shown in **1**E to form a plurality of integrated antenna package structures **100** shown in FIG. **1**F. The singulation process at least cuts the circuit structure **120**, the encapsulant **174**, and/or the dielectric body **177**.
(40) It is worth noting that after the singulation process is performed, the singulated elements still adopt similar reference numbers. For example, the chip **110** (shown in FIG. **1**E) may be the chip **110** after singulation (shown in FIG. **1**F); the circuit structure **120** (shown in FIG. **1**E) may be the circuit structure **120** after singulation (as shown in FIG. **1**F); the encapsulant **174** (shown in FIG. **1**E) may be the encapsulant **174** after singulation (shown in FIG. **1**F); and the dielectric body **177** (shown in FIG. **1**E) may be the dielectric body **177** after singulation (shown in FIG. **1**F). The description for the singulated films, layers, or components with the same reference number is not repeated or specifically shown hereinafter.
(41) It should be noted that, in this embodiment, the carrier **91** is removed first, before a singulation process or other suitable processes for forming the integrated antenna package structures **100** are performed. In an embodiment not shown, the singulation process is performed on the structures on the carrier **91** first, and after the carrier **91** is removed, the integrated antenna package structures **100** are then formed.
(42) As shown in FIG. **1**F, in this embodiment, after the carrier **91** is removed, a plurality of conductive terminals **180** are formed on the circuit structure **120** to form the corresponding integrated antenna package structure **100**.
(43) In one embodiment, the carrier **91** is removed first, and a plurality of conductive terminals **180**

are then formed, and then a singulation process is performed. In one embodiment, the singulation process is performed first, then the carrier **91** is removed, and then the conductive terminals **180** are formed.
(44) After the above processes, the fabrication of the integrated antenna package structure **100** of this embodiment may be considered substantially completed.
(45) FIG. **1**F is a partial cross-sectional schematic diagram of a package structure of a chip **110** according to the first embodiment of the disclosure. FIG. **1**G is a schematic partial top view of a package structure of a chip **110** according to the first embodiment of the disclosure. For example, FIG. **1**F may be a schematic cross-sectional view corresponding to the line I-I′ in FIG. **1**G. In addition, for the sake of clarity, only part of the film layers or members is shown in FIG. **1**G. For example, FIG. **1**G only shows exemplarily the dielectric body **177**, the shielding layer **165**, or the second antenna layer **166**; and, the chip **110**, the first conductive portion **134**, the conductive member **142**, or the support member **148** are also shown in perspective.
(46) Please refer to FIG. **1**F and FIG. **1**G. The integrated antenna package structure **100** includes a chip **110**, a circuit structure **120**, a shielding body **130**, an encapsulant **174**, a first antenna layer **156**, a dielectric body **177**, and a second antenna layer **166**. The chip **110** is electrically connected to the corresponding circuit in the circuit structure **120**. The shielding body **130** is disposed on the circuit structure **120**. The shielding body **130** has an accommodating space **131**. The chip **110** is disposed in the accommodating space **131** of the shielding body **130**. The encapsulant **174** is disposed on the circuit structure **120** and covers the chip **110**. The first antenna layer **156** is disposed on the circuit structure **120**. The first antenna layer **156** may be further disposed on the encapsulant **174**. Antennas in the first antenna layer **156** are electrically connected to corresponding circuits in the circuit structure **120**. The dielectric body **177** is disposed on the second sealing surface **174***b* of the encapsulant **174**. The second antenna layer **166** is on the dielectric body **177**.
(47) In this embodiment, in the thickness direction of the integrated antenna package structure **100** (e.g., the normal direction of the sealing surface **174***a* or the sealing surface **174***b*), the second antenna layer **166** overlaps the first antenna layer **156**, and the dielectric body **177** is disposed between the first antenna layer **156** and the second antenna layer **166**.
(48) In this embodiment, the shielding body **130** includes a first conductive portion **134** and a second conductive portion **135**. The first conductive portion **134** and the second conductive portion **135** constitute the accommodating space **131**. The first conductive portion **134** penetrates through the encapsulant **174** and surrounds the chip **110**. The second conductive portion **135** is disposed on the second sealing surface **174***b* of the encapsulant **174**. The second conductive portion **135** overlaps the chip **110** in a direction perpendicular to the first sealing surface **174***a* or the second sealing surface **174***b* (as shown in FIG. **1**G). In this way, the interference between the chip **110** and the antenna may be reduced.
(49) In one embodiment, the first conductive portion **134**, the second conductive portion **135**, and the circuit structure **120** constitute a closed accommodating space **131**. That is to say, the components (e.g., the conductive member **142**) for electrically connecting the circuit structure **120** and the first antenna layer **156** are disposed outside the closed accommodating space **131**. That is, the components (e.g., the conductive member **142**) for electrically connecting the circuit structure **120** and the first antenna layer **156** do not penetrate through the closed accommodating space **131**.
(50) In one embodiment, the shielding body **130** is grounded. Grounding includes a floating ground or a physical ground. For example, the shielding body **130** is grounded through a ground circuit and a ground terminal **183** (one of the conductive terminals **180**) in the circuit structure **120**.
(51) In one embodiment, the projection of the first conductive portion **134** on the first sealing surface **174***a* or the second sealing surface **174***b* is within the projection of the second conductive portion **135** on the first sealing surface **174***a* or the second sealing surface **174***b* to provide a better process window.
(52) In this embodiment, the integrated antenna package structure **100** further includes a pillar-

shaped conductive connector **142**. The conductive connector **142** penetrates through the encapsulant **174**. Antennas in the first antenna layer **156** are electrically connected to corresponding circuits in the circuit structure **120** through corresponding conductive connectors **142**.
(53) In one embodiment, the first conductive portion **134** of the shielding body **130** and the conductive connector **142** are formed by the same process. In other words, the materials of the first conductive portion **134** and the conductive connector **142** are substantially the same, and the thickness of the first conductive portion **134** and that of the conductive connector **142** are also substantially the same.
(54) In one embodiment, if the first antenna layer **156** has a plurality of antennas that are electrically separated from each other, the shielding body **130** is disposed between those antennas in a direction parallel to the first sealing surface **174***a* or the second sealing surface **174***b*. For example, the second conductive portion **135** of the shielding body **130** is disposed between those antennas. In this way, the interference between the antennas may be reduced, thereby improving the application of the antenna-integrated package structure **100** and/or the corresponding signal quality.
(55) In one embodiment, if the first antenna layer **156** has a plurality of antennas that are electrically separated from each other, the shielding body **130** is disposed between the conductive connectors **142** for connecting the antennas in a direction parallel to the first sealing surface **174***a* or the second sealing surface **174***b*. For example, the first conductive portion **134** of the shielding body **130** may be disposed between the conductive connectors **142** for connecting the antennas. For another example, the shielding body **130** may be disposed between the conductive connector **142***a* and the conductive connector **142***b*, the shielding body **130** is disposed between the conductive connector **142***b* and the conductive connector **142***c*, and/or the shielding body **130** is disposed between the conductive connector **142***a* and the conductive connector **142***c*. In this way, the interference between the antennas may be reduced, thereby improving the application of the antenna-integrated package structure **100** and/or the corresponding signal quality.
(56) In one embodiment, if the first antenna layer **156** has a plurality of antennas that are electrically separated from each other, the shielding body **130** is disposed between the conductive connectors **142** for connecting different two antennas in any direction parallel to the first sealing surface **174***a* or the second sealing surface **174***b*.
(57) In this embodiment, the antenna pattern of the second antenna layer **166** corresponds to the antenna pattern of the first antenna layer **156**. That is to say, in a direction (e.g., the thickness direction) perpendicular to the first sealing surface **174***a* or the second sealing surface **174***b*, the antenna pattern of the second antenna layer **166** does not overlap the chip **110**; or further, the antenna pattern of the second antenna layer **166** does not overlap the shielding body **130**. As such, the interference between the antenna and the chip **110** may be reduced, thereby improving the application of the antenna-integrated package structure **100** and/or the corresponding signal quality.
(58) In one embodiment, antennas in the first antenna layer **156** are coupled with the corresponding antennas in the second antenna layer **166**, and the coupled antennas are not electrically connected. In an exemplary application, an antenna corresponding to the first antenna layer **156** is a driven antenna, and a corresponding antenna in the second antenna layer **166** is a parasitic antenna, but the disclosure is not limited thereto.
(59) In this embodiment, the integrated antenna package structure **100** further includes a shielding layer **165**. The shielding layer **165** is disposed on the dielectric body **177**. The shielding layer **165** overlaps the chip **110** in a direction perpendicular to the first sealing surface **174***a* or the second sealing surface **174***b*. In this way, the interference of external electromagnetic waves to the chip **110** may be reduced.
(60) In one embodiment, the pattern of the shielding layer **165** overlaps or corresponds to the pattern of the second conductive portion **135**, which improves the shielding quality of the shielding layer **165**.
(61) In one embodiment, as shown in FIG. **1**G, the projection of the chip **110** on the first sealing

surface **174***a* or the second sealing surface **174***b* is within the projection of the shielding layer **165** on the first sealing surface **174***a* or the second sealing surface **174***b*, and the projection of the chip **110** on the first sealing surface **174***a* or the second sealing surface **174***b* completely overlaps the projection of the shielding layer **165** on the first sealing surface **174***a* or the second sealing surface **174***b*. In this way, the interference of external electromagnetic waves to the chip **110** may be reduced.

(62) In one embodiment, if the first antenna layer **156** or the second antenna layer **166** has multiple antennas that are electrically separated from each other (e.g., driven antennas that are electrically separated from each other; or, parasitic antennas that are electrically separated from each other); and the shielding layer **165** is disposed between the antennas formed by the second antenna layer **166** in a direction parallel to the first sealing surface **174***a* or the second sealing surface **174***b*. In this way, the interference between the antennas may be reduced, thereby improving the application of the antenna-integrated package structure **100** and/or the corresponding signal quality.

(63) In one embodiment, if the first antenna layer **156** or the second antenna layer **166** has multiple antennas that are electrically separated from each other (e.g., driven antennas that are electrically separated from each other; or, parasitic antennas electrically separated from each other), the shielding layer **165** is disposed between the antennas formed by the second antenna layer **166** in any direction parallel to the first sealing surface **174***a* or the second sealing surface **174***b*.

(64) In one embodiment, the shielding layer **165** is grounded, but the disclosure is not limited thereto. For example, the grounding of the shielding layer **165** is floating.

(65) In one embodiment, in terms of electrical properties, the shielding layer **165** and the second conductive portion **135** are not electrically connected to each other. In one embodiment, there is no conductive substance between the shielding layer **165** and the second conductive portion **135** structurally.

(66) In one embodiment, the width (i.e., another dimension perpendicular to the direction in which the shielding layer **165** has the largest dimension) of the shielding layer **165** is larger than the width of any antenna in the second antenna layer **166** (i.e., the dimension perpendicular to the extending direction of the antenna). As such, with the relatively simple pattern design mentioned above (e.g., adjusting the width of the shielding layer **165**), other components and/or processes may be omitted (for example, the shielding layer **165** does not need to be electrically connected to other conductors in the integrated antenna package structure **100** to be grounded), but the overall charge capacity of the shielding layer **165** may still be improved, such that the shielding layer **165** has a good shielding effect.

(67) In one embodiment, the projection area of the shielding layer **165** on the surface of the dielectric body **177** is larger than the projection area of the second antenna layer **166** on the surface of the dielectric body **177**, thereby improving the overall charge capacity of the shielding layer **165**, and the shielding layer **165** has a good shielding effect.

(68) In this embodiment, the integrated antenna package structure **100** further includes a support member **148**. The support member **148** may penetrate through the encapsulant **174**. The material, appearance, and/or manufacturing technique of the support member **148** may be the same as those of the conductive connector **142**; however, in this embodiment, the support member **148** does not overlap the first antenna layer **156** and/or the second antenna layer **166** in a direction perpendicular to the first sealing surface **174***a* or the second sealing surface **174***b*.

(69) The arrangement or the quantity of the support members **148** may be adjusted according to the design requirements, which is not limited in the disclosure. For example, the support members **148** may be disposed where the conductive connectors **142** are relatively sparse, such that the corresponding structure or stress bearing may be well balanced in the manufacturing process and/or application of the integrated antenna package structure **100**.

(70) In one embodiment, the support member **148** is a dummy component for signal processing or signal transmission of the integrated antenna package structure **100**. In other words, the support

member **148** may substantially not participate in signal processing or signal transmission.
(71) Note that the disclosure does not limit the support member **148** to not be electrically connected to any conductor. For example, in an embodiment not directly shown, the support member **148** is electrically connected to the shielding body **130** through a corresponding circuit in the circuit structure **120**, as a way to improve the overall charge capacity of the shielding body **130** and the conductors (e.g., the conductive member **140** and/or the support member **148**) electrically connected thereto.
(72) FIG. **2** is a partial cross-sectional schematic diagram of a partial manufacturing method of an integrated antenna package structure according to a second embodiment of the disclosure. The manufacturing method of the integrated antenna package structure **200** of the present embodiment is similar to the manufacturing method of the integrated antenna package structure **100**. As similar components with the same reference numbers have similar functions, materials, or forming techniques, the description thereof is omitted hereinafter.
(73) As shown in FIG. **2**, different from the integrated antenna package structure **100**, the integrated antenna package structure **200** does not have a shielding layer the same as or similar to the shielding layer **165** described above (since there is no shielding layer, it is not shown or marked).
(74) FIG. **3** is a partial cross-sectional schematic diagram of a partial manufacturing method of an integrated antenna package structure according to a third embodiment of the disclosure. The manufacturing method of the integrated antenna package structure **300** of the present embodiment is similar to the manufacturing method of the integrated antenna package structure **100**. As similar components with the same reference numbers have similar functions, materials, or forming techniques, the description thereof is omitted hereinafter.
(75) As shown in FIG. **3**, the integrated antenna package structure **300** includes a chip **110**, a circuit structure **120**, a shielding body **130**, an encapsulant **174**, a first antenna layer **356**, a dielectric body **177**, and a second antenna layer **166**. The first antenna layer **356** is disposed on the circuit structure **120**. Antennas in the first antenna layer **356** are electrically connected to corresponding circuits in the circuit structure **120**.
(76) In this embodiment, the encapsulant **174** is disposed between the first antenna layer **356** and the dielectric body **177**.
(77) In one embodiment, the topmost insulating layer **122** (e.g., an insulating layer **122** closest to the encapsulant **174**) in the circuit structure **120** has a plurality of openings, and part of the topmost conductive layer **123** is embedded in the openings. Part of the topmost conductive layer **123** (e.g., a conductive layer **121** closest to the encapsulant **174**) in the circuit structure **120** constitutes the first antenna layer **356**, but the disclosure is not limited thereto.
(78) To sum up, the integrated antenna package structure and the manufacturing method thereof of the disclosure have small volume and/or good quality.
(79) Finally, it should be noted that the above embodiments serve only to illustrate the technical solutions of the disclosure, and not to limit them. Although the disclosure has been described in detail with reference to these embodiments, those of ordinary skill in the art should understand that it is still possible to modify the technical solutions provided in the embodiments, or to perform equivalent replacements for some or all of the technical features. However, these modifications or substitutions do not deviate the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of the disclosure.

Claims

1. An integrated antenna package structure, comprising: a chip; a circuit structure electrically connected to the chip; a shielding body disposed on the circuit structure and having an accommodating space, wherein the chip is disposed in the accommodating space of the shielding body; an encapsulant disposed on the circuit structure and covering the chip; a first antenna layer

disposed on the circuit structure and electrically connected to the circuit structure; a dielectric body disposed on the encapsulant; a second antenna layer disposed on the dielectric body; and a conductive connector penetrating through the encapsulant, wherein the shielding body comprises: a first conductive portion penetrating through the encapsulant and surrounding the chip; and a second conductive portion disposed on the encapsulant and overlapping the chip, and wherein: the first conductive portion is electrically connected to the second conductive portion, and the first conductive portion and the second conductive portion constitute the accommodating space; the first antenna layer is electrically connected to the circuit structure by the conductive connector; and a material and a thickness of the first conductive portion are same as a material and a thickness of the conductive connector.

2. The integrated antenna package structure according to claim 1, wherein the first antenna layer and the second conductive portion are from a same film layer.

3. The integrated antenna package structure according to claim 1, further comprising: a support member penetrating through the encapsulant, wherein a material and a thickness of the support member are same as the material and the thickness of the conductive connector.

4. The integrated antenna package structure according to claim 1, further comprising: a shielding layer disposed on the dielectric body and overlapping the chip, wherein the shielding layer and the second antenna layer are from a same film layer.

5. The integrated antenna package structure according to claim 4, wherein the second antenna layer comprises a first antenna area and a second antenna area that are electrically separated from each other, and the shielding layer is disposed between the first antenna area and the second antenna area.

6. The integrated antenna package structure according to claim 4, wherein the shielding layer is grounded.

7. A manufacturing method of an integrated antenna package structure, the manufacturing method comprising: forming a circuit structure; forming a conductive connector on the circuit structure; disposing a chip on the circuit structure to be electrically connected with the circuit structure; forming a shielding body on the circuit structure, wherein the shielding body has an accommodating space, and the chip is disposed in the accommodating space of the shielding body; forming an encapsulant on the circuit structure to cover the chip; forming a first antenna layer on the circuit structure to be electrically connected with the circuit structure; forming a dielectric body on the encapsulant; and forming a second antenna layer on the dielectric body, wherein the shielding body comprises: a first conductive portion penetrating through the encapsulant and surrounding the chip; and a second conductive portion disposed on the encapsulant and overlapping the chip, and wherein: the conductive connector penetrates through the encapsulant; the first conductive portion is electrically connected to the second conductive portion, and the first conductive portion and the second conductive portion constitute the accommodating space; the first antenna layer is electrically connected to the circuit structure by the conductive connector; and a material and a thickness of the first conductive portion are same as a material and a thickness of the conductive connector.

8. The manufacturing method of the integrated antenna package structure according to claim 7, wherein forming the shielding body and forming the encapsulant comprise: forming the first conductive portion on the circuit structure; forming the encapsulant on the circuit structure to cover the chip and laterally cover the first conductive portion; forming the second conductive portion on the encapsulant that overlaps the chip.

9. The manufacturing method of the integrated antenna package structure according to claim 8, wherein the first antenna layer and the second conductive portion are formed by same processes.

10. The manufacturing method of the integrated antenna package structure according to claim 8, wherein after the encapsulant is formed, the encapsulant further laterally covers the conductive connector; after the first antenna layer is formed, the first antenna layer is electrically connected to

the circuit structure through the conductive connector; and the conductive connector and the first conductive portion are formed by same processes.

11. The manufacturing method of the integrated antenna package structure according to claim 10, further comprising: forming a support member on the circuit structure, wherein after the encapsulant is formed, the encapsulant further laterally covers the support member; and the support member and the conductive connector are formed by same processes.

12. The manufacturing method of the integrated antenna package structure according to claim 7, wherein forming the first antenna layer precedes forming the encapsulant.

13. The manufacturing method of the integrated antenna package structure according to claim 7, further comprising: forming a shielding layer on the dielectric body to overlap the chip, wherein the shielding layer and the second antenna layer are formed by same processes.

14. The integrated antenna package structure according to claim 1, wherein a portion of the encapsulant is disposed between the chip and the shielding body.

15. The integrated antenna package structure according to claim 3, wherein: the support member does not overlap the first antenna layer; the support member does not overlap the second antenna layer; the support member is a dummy component for signal processing or signal transmission; or the support member is electrically connected to the shielding body.

16. An integrated antenna package structure, comprising: a chip; a circuit structure electrically connected to the chip; a shielding body disposed on the circuit structure and having an accommodating space, wherein the chip is disposed in the accommodating space of the shielding body; an encapsulant disposed on the circuit structure and covering the chip; a first antenna layer disposed on the circuit structure and electrically connected to the circuit structure; a dielectric body disposed on the encapsulant; and a second antenna layer disposed on the dielectric body, wherein the encapsulant is disposed between the first antenna layer and the dielectric body.

17. The integrated antenna package structure according to claim 16, wherein the shielding body comprises: a first conductive portion penetrating through the encapsulant and surrounding the chip; and a second conductive portion disposed on the encapsulant and overlapping the chip, wherein the first conductive portion is electrically connected to the second conductive portion, and the first conductive portion and the second conductive portion constitute the accommodating space.